Measurement Conversions

A big unit holds many small units. Multiply to change units.

1 meter is 100 centimeters

$3\ \text{m} = 3 \times 100 = 300\ \text{cm}$

Big unit to small unit: multiply.

Make a table

ft	in
1	12
2	24
3	36
4	48
5	60

1 foot = 12 inches.
Each row adds 12 more.
5 ft = 5 × 12 = 60 in.

Units to know

1 km = 1,000 m 1 m = 100 cm 1 kg = 1,000 g 1 L = 1,000 mL
1 yd = 3 ft 1 ft = 12 in 1 lb = 16 oz
1 hr = 60 min 1 min = 60 sec 1 day = 24 hr 1 week = 7 days

Mixed units

$2\ \text{m}\ 35\ \text{cm} = 200\ \text{cm} + 35\ \text{cm} = 235\ \text{cm}$

Your turn!

$4\ \text{ft} = \square\ \text{in}$

Measurement Conversions: Practice

Big unit to small unit: multiply by how many fit in one.

Part 1 Fill in the table The first row is done.

1)

m	cm
1	100
2	
4	
6	
9	

2)

yd	ft
1	3
3	
5	
7	
10	

3)

lb	oz
1	16
2	
3	
5	
8	

4)

hr	min
1	60
2	
4	
5	
10	

Part 2 Metric units No help this time.

5) 4 m = ____ cm

6) 3 kg = ____ g

7) 5 L = ____ mL

8) 2 km = ____ m

9) 7 m = ____ cm

10) 9 kg = ____ g

Measurement Conversions: Practice (continued)

Part 2 Metric units (continued)

11) 6 km = ☐ m

12) 8 L = ☐ mL

13) 12 m = ☐ cm

14) 4 kg = ☐ g

15) 3 L = ☐ mL

16) 10 km = ☐ m

Part 3 Feet, pounds and time No help this time.

17) 3 ft = ☐ in

18) 4 yd = ☐ ft

19) 2 lb = ☐ oz

20) 5 min = ☐ sec

21) 3 hr = ☐ min

22) 2 yd = ☐ in

Measurement Conversions: Practice (continued)

Part 3 Feet, pounds and time (continued)

23) 6 ft = ☐ in

24) 3 weeks = ☐ days

25) 4 lb = ☐ oz

26) 2 days = ☐ hr

27) 8 yd = ☐ ft

28) 7 min = ☐ sec

Part 4 Mixed units No help this time. Change the big unit, then add.

29) 3 m 42 cm = ☐ cm

30) 1 kg 250 g = ☐ g

31) 3 ft 4 in = ☐ in

32) 2 hr 15 min = ☐ min

33) 4 m 8 cm = ☐ cm

34) 2 L 600 mL = ☐ mL

Measurement Conversions: Practice (continued)

Part 4 Mixed units (continued)

35) 5 ft 6 in = ☐ in

36) 1 hr 45 min = ☐ min

37) 2 yd 1 ft = ☐ ft

38) 3 lb 5 oz = ☐ oz

Part 5 Which is more? No help this time. Change to the same unit first.

39) 3 ft ☐ 40 in

40) 2 m ☐ 150 cm

41) 1 kg ☐ 1,200 g

42) 5 yd ☐ 14 ft

43) 2 hr ☐ 100 min

44) 3 L ☐ 3,000 mL

45) 2 lb ☐ 40 oz

46) 4 min ☐ 200 sec

Measurement Conversions: Practice (continued)

Part 6 Add and take away No help this time. Use the small unit.

47) 1 L − 350 mL = ☐ mL

48) 2 m + 45 cm = ☐ cm

49) 3 ft − 8 in = ☐ in

50) 1 hr + 25 min = ☐ min

51) 1 kg − 375 g = ☐ g

52) 2 yd + 2 ft = ☐ ft

53) 1 km − 400 m = ☐ m

54) 1 lb + 9 oz = ☐ oz

Part 7 Convert

55) 4 min = ☐ sec

56) 8 km = ☐ m

57) 10 hr = ☐ min

Measurement Conversions: Practice (continued)

Part 7 Convert (continued)

58) 3 lb = ____ oz

59) 8 min = ____ sec

60) 12 hr = ____ min

61) 7 L = ____ mL

62) 8 yd = ____ in

63) 3 km = ____ m

64) 8 lb = ____ oz

65) 4 weeks = ____ days

66) 11 m = ____ cm

67) 10 yd = ____ ft

68) 11 L = ____ mL

69) 5 yd = ____ in

Measurement Conversions: Practice (continued)

Part 7 Convert (continued)

70) 11 weeks = ☐ days

71) 12 ft = ☐ in

72) 3 m = ☐ cm

73) 9 lb = ☐ oz

74) 3 min = ☐ sec

75) 9 min = ☐ sec

76) 2 hr = ☐ min

77) 9 ft = ☐ in

78) 11 min = ☐ sec

79) 3 yd = ☐ ft

80) 9 yd = ☐ in

81) 11 kg = ☐ g

Measurement Conversions: Practice (continued)

Part 7 Convert (continued)

82) 8 kg = ☐ g

83) 11 hr = ☐ min

84) 4 hr = ☐ min

85) 8 m = ☐ cm

86) 11 days = ☐ hr

87) 5 yd = ☐ ft

88) 11 yd = ☐ in

89) 9 hr = ☐ min

90) 3 yd = ☐ in

Part 8 Remember? Problems from earlier packets.

91) 0.7 ☐ 0.65

92) $\frac{38}{100}$ = ☐

Measurement Conversions: Practice (continued)

Part 8 Remember? (continued)

93) $5 \times \frac{3}{4} = \frac{\square}{4}$

94) $2\frac{5}{6} + 1\frac{3}{6} = \square\frac{\square}{6}$

95) $36 \times 25 = \square$

96) $4{,}568 \div 8 = \square$

Part 9 Word problems Change to the same unit first.

97) A rope is 3 m long. Leo cuts off 75 cm. How long is the rope now, in centimeters?

Number sentence _____

Answer ☐ cm

Draw or show your work

98) A movie is 2 hours long. Ana has watched 85 minutes. How many minutes are left?

Number sentence _____

Answer ☐ minutes

Draw or show your work

99) A bag of apples weighs 2 kg. Each apple is about 200 g. About how many apples are in the bag?

Number sentence _____

Answer ☐ apples

Draw or show your work

100) Mia drinks 250 mL of milk each day. How much does she drink in a week?

Number sentence _____

Answer ☐ mL

Draw or show your work

Measurement Conversions: Practice (continued)

Part 9 Word problems (continued)

101) A desk is 4 ft long. A shelf is 50 in long. Which is longer?

Number sentence _____

Answer

Draw or show your work

102) Challenge: Ben runs 400 m around the track 6 times. Is that more than 2 km?

Number sentence _____

Answer

Draw or show your work

Check-In: Measurement Conversions

Show what you know. No help this time. Score: _____ / 9

Part 1 Fill in the table

1)

min	sec
1	60
2	
3	
6	
8	

Part 2 Change the unit

2) 7 yd = ____ in

3) 5 lb = ____ oz

4) 2 kg = ____ g

5) 5 hr = ____ min

6) 2 m = ____ cm

7) 12 days = ____ hr

Part 3 Mixed units

8) 4 m 5 cm = ____ cm

9) 6 ft 1 in = ____ in